EECE-5644 Machine Learning & Pattern Recognition 2026-SPRG

GOALS: Game Outcome and Analytics Learning System

Predicting Premier League Match Outcomes from Player Performance Data

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Source Code & Data Pipeline

https://github.com/natmaw46new/GOALS-Game_Outcome_and_Analytics_Learning_System-

ABSTRACT

This paper presents GOALS (Game Outcome and Analytics Learning System), multi-stage machine learning pipeline for predicting Premier League match outcomes from per-player performance statistics. Four seasons of FotMob data (20,000–30,000 player-match observations) processed through position-aware composite score construction, Ridge and Random Forest regression for player performance prediction, K-Means clustering for archetype discovery, and Logistic Regression and Random Forest classification for match outcome prediction (home-win, draw, away-win). A temporal train/test split prevents data leakage. The best regression model achieves position-grouped test R^2 of 0.938–0.966; the best classifier achieves a macro-averaged F1 of 0.542, exceeding the majority-class baseline of 0.207.

CONTENTS

I. Introduction . . . 4
A. Motivation and Problem Statement . . . 4
B. Related Work . . . 4
C. Specific Contributions . . . 5
D. Report Structure . . . 5
II. Data Description . . . 5
A. Data Source and Collection . . . 5
B. Dataset Characteristics . . . 6
C. Exploratory Data Analysis . . . 7
1. Feature Distributions . . . 7
2. Correlation Structure . . . 8
3. PCA Projection . . . 9
4. Composite Score Validation . . . 10
III. Methodology . . . 11
A. Pipeline Overview . . . 11
B. Preprocessing and Feature Engineering . . . 12
1. Temporal Train/Test Split . . . 12
2. Ratio Features and Imputation . . . 12
3. Rolling Feature Windows . . . 12
4. Position-Aware Composite Performance Scores . . . 12
5. Sensitivity Analysis . . . 14
6. Scaling . . . 14
C. Regression: Player Performance Prediction . . . 14
1. Problem Formulation . . . 14
2. Ridge Regression . . . 14
3. Random Forest Regression . . . 15
4. Evaluation . . . 15
D. Clustering: Player Archetype Discovery . . . 15

1. Pipeline per Position Group . . . 15
2. Dimensionality Reduction . . . 15
3. Number of Clusters . . . 15
E. Classification: Match Outcome Prediction . . . 16
1. Team-Level Aggregation . . . 16
2. Differential Features . . . 16
3. Models and Evaluation . . . 16
IV. Results . . . 17
A. Regression Results . . . 17
1. Baseline and Ridge Regression . . . 17
2. Position-Grouped Random Forest and Feature Importance . . . 17
B. Clustering Results . . . 19
1. PCA Variance and Model Selection . . . 19
2. Archetype Profiles . . . 19
3. Archetype Heatmap . . . 20
4. PCA Scatter . . . 21
C. Classification Results . . . 21
1. Cross-Validation and Test Performance . . . 21
2. Confusion Matrix and Feature Importance . . . 23
V. Discussion . . . 24
A. Interpretation of Results . . . 24
1. Regression . . . 24
2. Clustering . . . 24
3. Classification . . . 24
B. Why Certain Methods Performed Better . . . 25
C. Lessons Learned . . . 25
D. Future Improvements . . . 25
VI. Conclusions . . . 26
Vita . . . 28
Appendix . . . 29

I. Introduction

A. Motivation and Problem Statement

Football analytics has undergone a quiet revolution over the past decade. The widespread adoption of event-level tracking data—expected goals (xG), progressive passes, pressures applied—has transformed the vocabulary of the sport at every level from elite clubs to broadcast punditry [3]. Yet the truly actionable output of this data revolution—a principled, interpretable system for forecasting what will happen in the *next* match—remains largely confined to proprietary club analytics departments.

This project sets out to close that gap. GOALS constructs an end-to-end machine learning pipeline that ingests publicly available per-player match statistics, builds position-aware performance representations, discovers data-driven player archetypes, and ultimately delivers probabilistic forecasts of Premier League match outcomes. The central prediction task is framed as a three-class classification problem: given the recent form of the players selected by two opposing managers, will the match result in a home win (**H**), a draw (**D**), or an away win (**A**)?

B. Related Work

Machine learning approaches to football outcome prediction fall broadly into three families. The earliest work used betting-market odds and historical result sequences as features [6]. A second generation incorporated team-level statistics using logistic regression and naive Bayes [7]. More recent work has moved to player-level representations: [8] introduced the *PlayeRank* framework aggregating per-event player ratings into team-level predictions, and [9] demonstrated that expected-goal-based features outperform raw goal counts for outcome classification.

The present work sits in this third family but contributes three distinctions: (i) a position-specific composite score combining FBref and FotMob metrics with domain-motivated weights and sensitivity analysis; (ii) a K-Means clustering stage producing interpretable player archetypes; and (iii) a rigorous season-level temporal split preventing future information leakage.

C. Specific Contributions

1. A two-source data collection and alignment pipeline integrating FBref and FotMob across four Premier League seasons (2021/22–2024/25).

2. Position-aware composite performance scores for attackers, midfielders, defenders, and goalkeepers, validated against an independent FotMob player rating via sensitivity analysis.

3. Ridge and Random Forest regression models predicting future player composite scores, achieving test R^2 of 0.938–0.966 across position-grouped models.

4. K-Means clustering (per position group, PCA-reduced) identifying interpretable player archetypes confirmed by radar charts, heatmaps, and temporal stability analysis.

5. Logistic Regression and Random Forest classifiers predicting match outcomes from aggregated team performance features, with differential feature engineering encoding the relative quality gap between sides.

D. Report Structure

Section II describes the dataset, exploratory analysis, and preprocessing. Section III details the methodology for all three modelling tasks. Section IV presents results and comparisons. Section V discusses findings, limitations, and future directions. Conclusions are given in Section VI. Supporting visualisations and hyperparameter tables are collected in the Appendix.

II. Data Description

A. Data Source and Collection

Data collection began with FBref [4], a widely-used football statistics platform that provides advanced per-match metrics such as expected goals (xG), expected assists (xA), and progressive passes. However, a thorough audit of FBref's per-player coverage revealed a critical gap: key passing and physical metrics—pass completion totals, accurate crosses, dribble counts, recovery actions, and position-specific defensive events—are either absent or inconsistently re-

ported at the individual match level. Because these features are central to the position-specific composite score formulas designed for this project, FBref alone was insufficient as a data backbone.

FotMob [5] was adopted as the primary—and ultimately sole—data source. FotMob provides a richer per-match player statistics feed that covers the full feature set required: diving saves, acted-as-sweeper counts, and high claims for goalkeepers; tackles won, interceptions, clearances, and aerial duel outcomes for defenders; chances created, successful dribbles, pass accuracy, and ball recoveries for midfielders and attackers. Crucially, FotMob also supplies a proprietary per-match player rating (`rating_title`) that serves as an independent validation benchmark for the composite performance scores constructed in this project. All four seasons of data were collected exclusively from FotMob using a custom two-pass scraper (described below).

Because FotMob partially blocks automated access via Cloudflare Turnstile, a two-pass scraping strategy was implemented: Pass 1 parses `__NEXT_DATA__` JSON embedded in the match page HTML (fast, no authentication required); Pass 2 falls back to the `/api/data/matchDetails` API endpoint with a browser session cookie for matches where Pass 1 returned empty player statistics. Player names and team names were normalised (e.g. *AFC Bournemouth* → *Bournemouth*) and date-based match keys were used to deduplicate records across the two passes.

B. Dataset Characteristics

Table 1 summarises the dataset scope.

Table 1: Dataset Summary

Property	**Value**
Scope	All 20 Premier League clubs
Seasons	2021/22, 2022/23, 2023/24 (train); 2024/25 (test)
Total matches	~1,520
Player-match obs.	~20,000–30,000 (after quality filter)
Features per obs.	~40 statistical metrics
Target	Match outcome: H (home win), D (draw), A (away win)

A quality filter retained only observations with at least 30 minutes played *and* a valid FotMob `rating_title`. Table 2 summarises the feature categories available after merging both sources.

Table 2: Feature Categories (FotMob)

Category	**Example Features**
Attacking	Goals, Shots on Target, xG, Dribbles Completed
Playmaking	Assists, Chances Created, xA, Progressive Passes
Passing	Pass Completion Rate, Through Balls, Crosses
Defensive	Tackles Won, Interceptions, Clearances, Aerial Duels Won
Goalkeeping	Saves, Diving Saves, Saves Inside Box, High Claims
Physical	Recoveries, Touches, Dispossessed
Contextual	Home/Away Indicator, Opponent Strength

C. Exploratory Data Analysis

1. Feature Distributions

Figure 1 shows kernel density estimates of twelve rolling features by position group. Defenders exhibit sharp, narrow peaks for `accurate_passes` and `defensive_actions`, reflecting consistency of defensive work-rate across matches. Forwards and wingers show right-skewed distributions for `expected_goals` and `ShotsOnTarget`, consistent with the rarity of high-xG opportunities. The `goals` and `assists` distributions are near-identical across positions due to the dominance of the zero-event mass. Additional EDA plots including composite score distributions and GK validation are provided in Appendix VI..

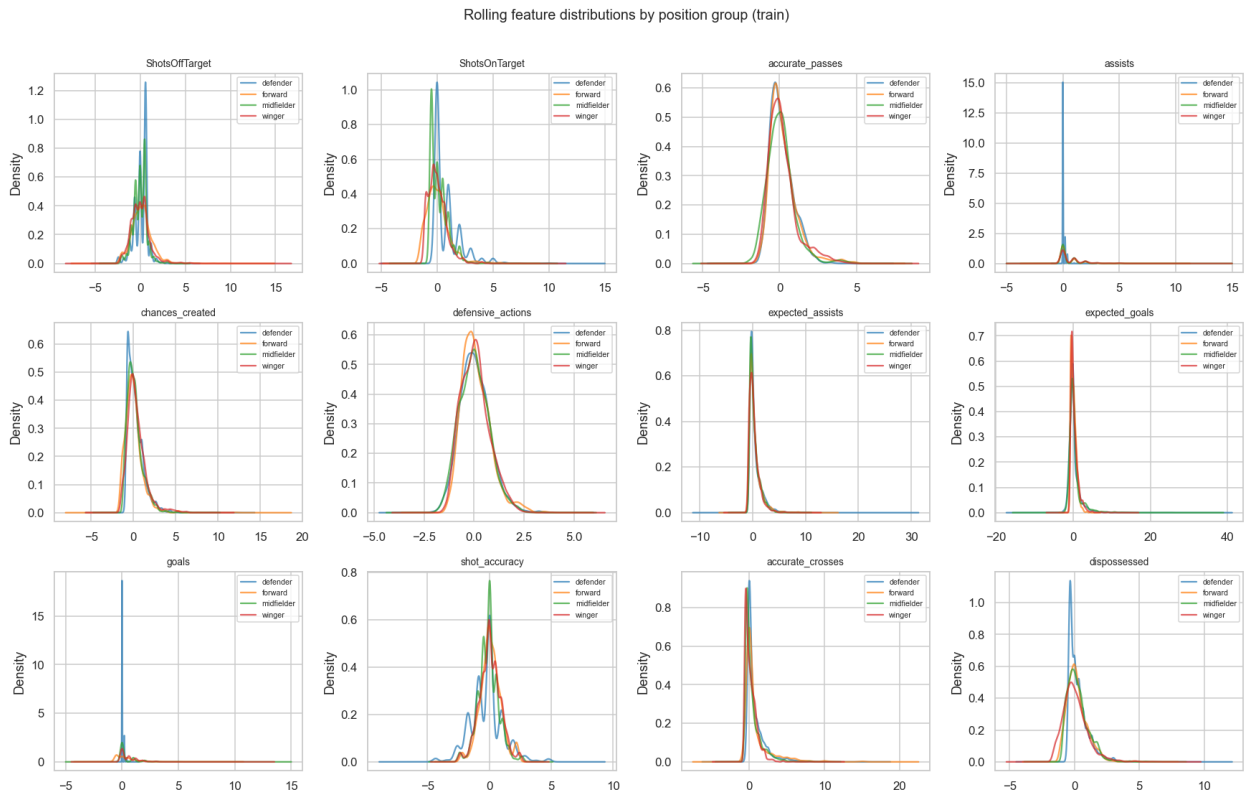

Figure 1: Kernel density estimates of 12 rolling features by position group (training data). Blue = defender, orange = forward, green = midfielder, red = winger.

2. Correlation Structure

Figure 2 shows the Pearson correlation matrix for the 20 features most strongly associated with the FotMob player rating. The strongest pairwise correlation is `roll5_ground_duels_won` ↔ `roll5_duel_won` ($r = 0.72$)—the only pair exceeding $|r| = 0.70$. The composite score correlates most strongly with `roll5_touches` ($r = 0.44$) and `roll5_accurate_passes` ($r = 0.37$), while the overall composite vs. `rating_title` correlation is $r = 0.19$.

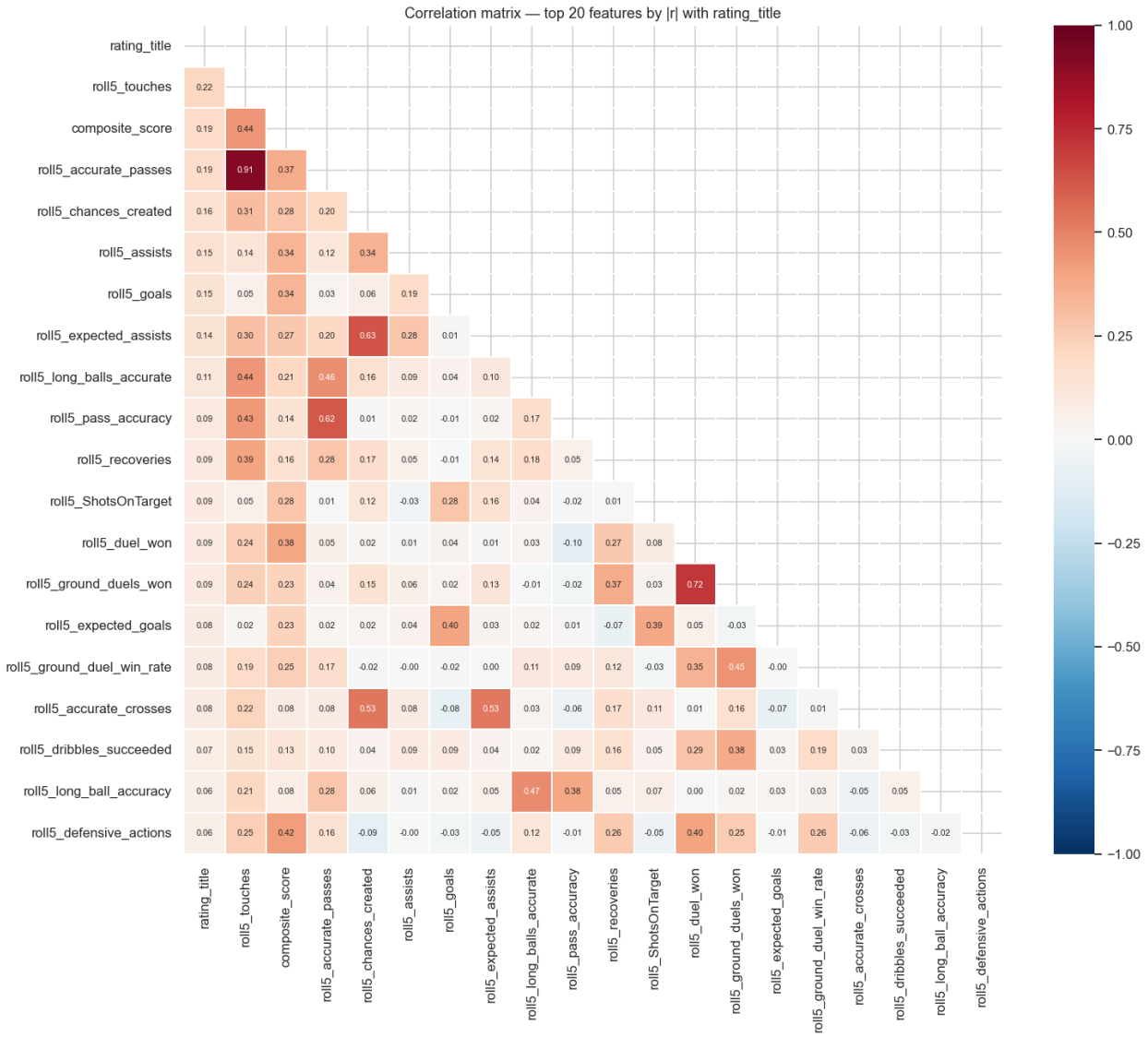

Figure 2: Pearson correlation matrix for the top 20 features by $|r|$ with `rating_title`. The one pair exceeding $|r| = 0.70$ is `roll5_ground_duels_won` ↔ `roll5_duel_won` ($r = 0.72$).

3. PCA Projection

Figure 3 shows a two-component PCA projection of all outfield training observations coloured by position group. PC1 (13.0% variance) separates high-volume attacking contributions (forwards and wingers extending along positive PC1) from defensive profiles (defenders concentrated near zero/negative PC1). PC2 (9.7% variance) captures vertical spread within the midfield cloud. The four position groups overlap substantially in PCA space, confirming that player

roles exist on a continuum.

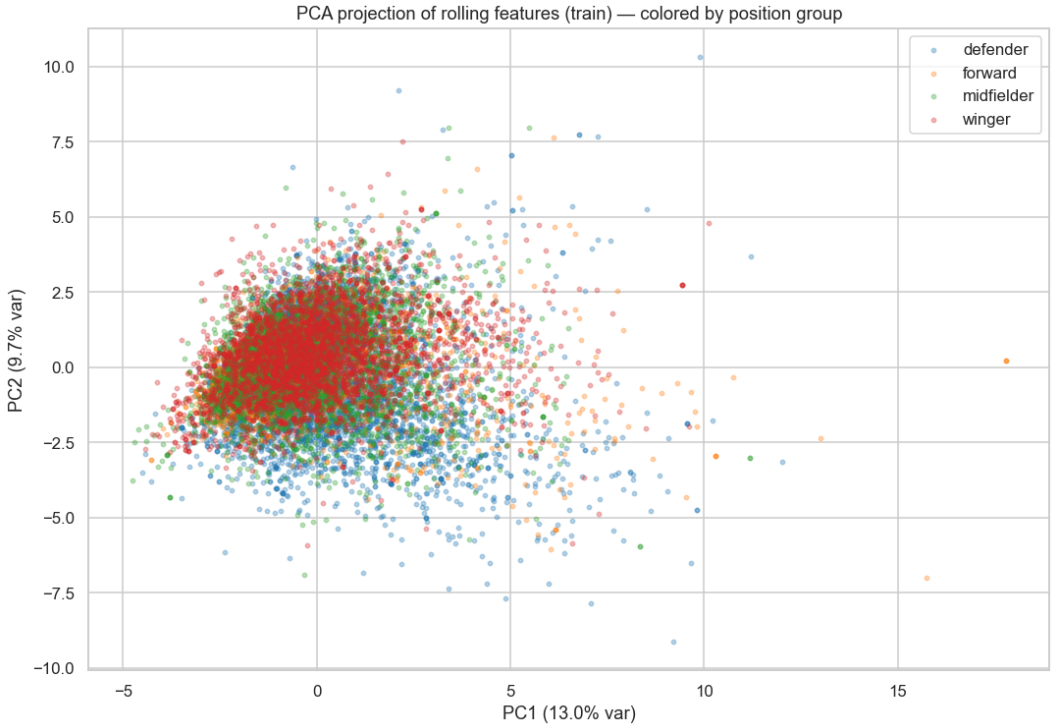

Figure 3: PCA projection of outfield rolling features onto two principal components coloured by position group. PC1 (13.0%) + PC2 (9.7%) together explain 22.7% of total variance.

4. Composite Score Validation

Figure 4 shows scatter plots of composite score versus FotMob `rating_title` per position group. Pearson correlations are $r = 0.066$ (defender, $n = 6{,}251$), $r = 0.302$ (midfielder, $n = 3{,}527$), $r = 0.292$ (winger, $n = 3{,}597$), and $r = 0.246$ (forward, $n = 1{,}913$). The low defender correlation is expected: FotMob's rating is influenced by ball-winning actions not proportionally captured by the defensive formula. The stronger correlations for midfielders, wingers, and forwards confirm the composite is meaningfully aligned with expert ratings for attacking-oriented roles. GK composite validation is provided in Appendix VI..

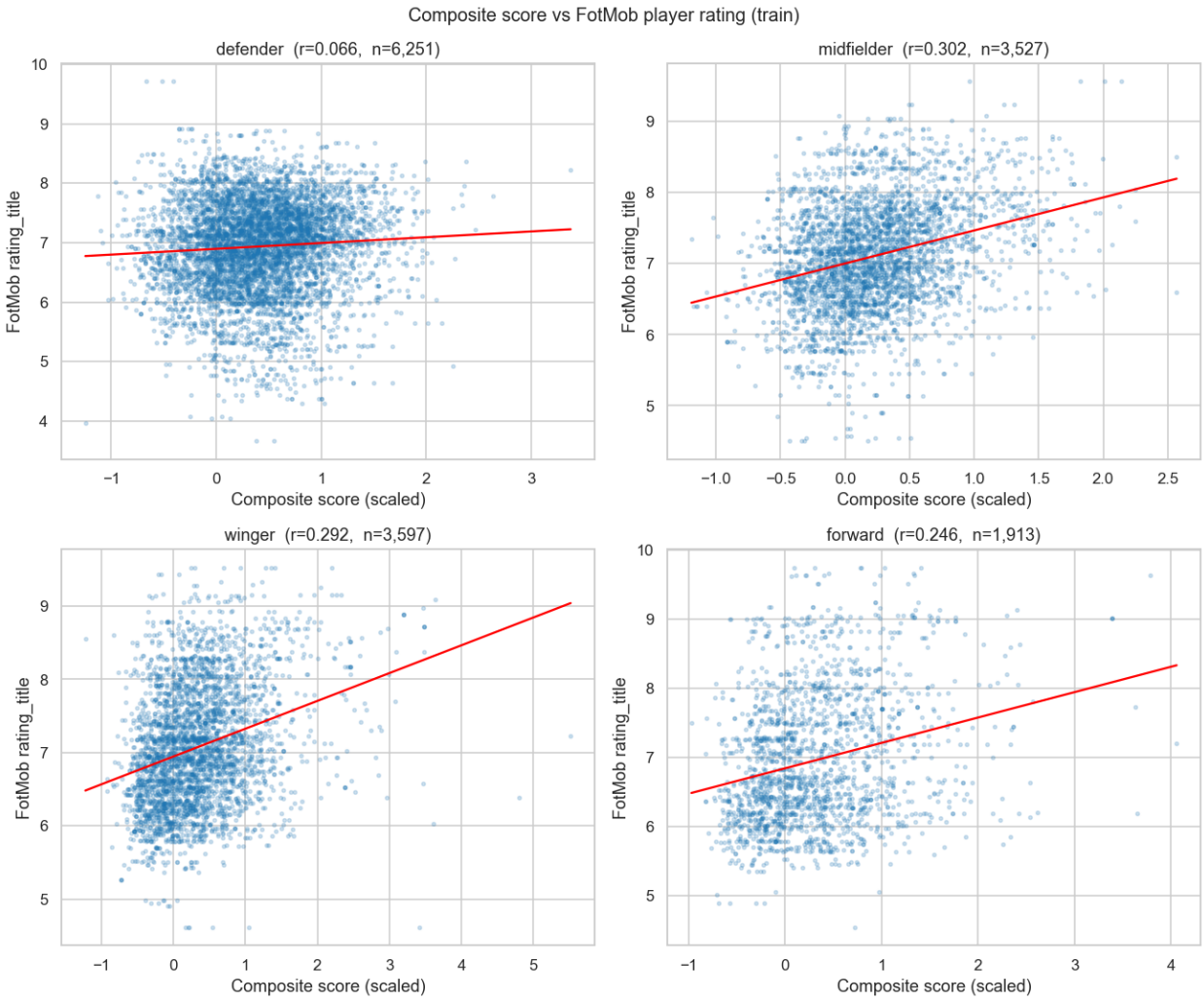

Figure 4: Composite score vs. FotMob `rating_title` per position group (training data). Red line: OLS trend. Pearson r: defender 0.066, midfielder 0.302, winger 0.292, forward 0.246.

III. Methodology

A. Pipeline Overview

The GOALS pipeline is a sequential, multi-stage system. Raw FotMob records flow through preprocessing and feature engineering, yielding per-player composite scores that feed into both a regression module and a clustering module. Aggregated team-level features from the regression module then enter the match outcome classifier. The five stages are: (1) data ingestion and preprocessing, (2) feature engineering and rolling aggregation, (3) composite score construction,

(4) player regression and clustering, and (5) team aggregation and classification.

B. Preprocessing and Feature Engineering

1. Temporal Train/Test Split

The three oldest seasons (2021/22–2023/24) form the training set and the most recent season (2024/25) the held-out test set, corresponding to an approximate 75/25 split (~1,140 training matches, ~380 test matches). A random shuffle was explicitly avoided as football results are inherently sequential.

Within training seasons, hyperparameter selection uses `TimeSeriesSplit` with five folds for regression and three folds for classification, walking forward chronologically across matchdays.

2. Ratio Features and Imputation

Fraction-based statistics (pass completion, aerial duel win rate, dribble success rate, shot accuracy) were computed as ratio = $n_{\text{success}}/n_{\text{total}}$, with zero-denominator entries filled using the training-set mean. Remaining numeric nulls were filled with training-set medians, fit on train only.

3. Rolling Feature Windows

All numeric statistics were converted to five-match rolling means using a `shift(1)` operation before the window:

$$x_{\text{roll}}^{(t)} = \frac{1}{\min(t-1,5)} \sum_{j=\max(1,t-5)}^{t-1} x^{(j)} \tag{1}$$

where $x^{(j)}$ is the raw statistic in the player's j-th chronological appearance. Observations with no prior appearances were dropped.

4. Position-Aware Composite Performance Scores

Four position-specific composite scores are defined. All inputs are first z-score normalised:

$$\tilde{x} = \frac{x-\mu}{\sigma} \tag{2}$$

then combined as a weighted sum:

$$S_{\text{pos}} = \sum_k w_k^{(\text{pos})} \cdot \tilde{x}_{k,\text{roll}} \tag{3}$$

where $\sum_k w_k^{(\text{pos})} = 1$ for each position. The full weight tables are given in Table 3.

Table 3: Position-Specific Composite Score Weighting Schemes

(a) Attacker / Winger

Metric	Src	w
Goals + Assists	FM	0.25
Expected Goals (xG)	FM	0.20
Expected Assists (xA)	FM	0.15
Successful Dribbles	FM	0.15
Total Shots	FM	0.10
Chances Created	FM	0.10
Ball Recoveries	FM	0.05

(b) Midfielder

Metric	Src	w
Progressive Passes	FM	0.20
Chances Created	FM	0.20
Expected Assists	FM	0.15
Goals + Assists	FM	0.15
Tackles Won	FM	0.15
Interceptions	FM	0.10
Ball Recoveries	FM	0.05

(c) Defender

Metric	Src	w
Tackles Won	FM	0.25
Aerial Duels Won	FM	0.20
Clearances	FM	0.20
Interceptions	FM	0.15
Blocks	FM	0.10
Progressive Passes	FM	0.10

(d) Goalkeeper

Metric	Src	w
Saves	FM	0.30
xG on Target Faced	FM	0.25
Diving Saves	FM	0.15
Saves Inside Box	FM	0.15
High Claims	FM	0.10
Acted as Sweeper	FM	0.05

FM = FotMob. w = weight. All weights sum to 1.00 per position.

5. Sensitivity Analysis

Three weight schemes were evaluated. Scheme A uses the proposed domain-motivated weights. Scheme B assigns equal weight to all metrics. Scheme C assigns weights proportional to $|r|$ of each rolling feature with `rating_title`. Results are summarised in Table 4.

Table 4: Sensitivity Analysis: Composite Score vs. FotMob Rating (Train)

Scheme	**Description**	r	**RMSE**
A (proposed)	Domain-motivated	0.188	1.021
B (equal)	Equal weights	0.163	1.038
C (data-driven)	Correlation-prop.	0.174	1.029

6. Scaling

A `RobustScaler` was fit per position group on training data and applied identically to the corresponding test subsets, reducing the influence of statistical outliers common in football match data.

C. Regression: Player Performance Prediction

1. Problem Formulation

The regression task predicts the composite score $S_i^{(t)}$ of player i in match t:

$$\hat{S}_i^{(t)} = f\left(\mathbf{x}_{\text{roll},i}^{(t)}\right) \tag{4}$$

Two feature sets were evaluated: *baseline* (rolling statistics plus home/away flag) and *enhanced* (baseline plus five-match rolling team attack and defensive strength proxies). Full hyperparameter search grids are given in Appendix VI..

2. Ridge Regression

Ridge regression solves:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha \|\boldsymbol{\beta}\|_2^2 \right\} \tag{5}$$

with $\alpha \in \{0.01, 0.1, 1, 10, 100, 500\}$ selected by `TimeSeriesSplit` cross-validation ($n = 5$).

3. Random Forest Regression

Random Forest regression aggregates B trees:

$$\hat{S} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}) \tag{6}$$

Hyperparameters were selected via `RandomizedSearchCV` (30 iterations, `TimeSeriesSplit` $n = 5$) over $B \in \{100, 200, 300\}$, max depth $\in \{4, 6, 8, 10, \text{None}\}$, min samples per leaf $\in [5, 30]$, and max features $\in \{\sqrt{p}, 0.5p\}$. Position-grouped models were trained separately for each outfield group.

4. Evaluation

Performance is measured by RMSE, MAE, and R^2:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{7}$$

The baseline predictor assigns each observation the training-set mean composite for its position group.

D. Clustering: Player Archetype Discovery

1. Pipeline per Position Group

K-Means clustering was applied independently to five position groups. Position-specific feature subsets (derived from the composite score formulas) were used; data were already `RobustScaler`-normalised. PCA variance curves per position group are provided in Appendix VI..

2. Dimensionality Reduction

PCA was applied before clustering to decorrelate the feature space. Components were chosen to retain approximately 90% cumulative variance: 10 components for outfield groups (defender 95%, midfielder 91%, winger 93%, forward 98%) and 8 for goalkeepers (86%).

3. Number of Clusters

k was swept from 2 to 8; $k = 3$ was chosen for all outfield positions for domain interpretability despite the silhouette peaking at $k = 2$. For goalkeepers, $k = 2$ was retained, producing a sub-

stantially higher silhouette score (≈ 0.63 vs. 0.13–0.21 outfield) that reflects the genuinely binary Shot-Stopper vs. Sweeper-Keeper distinction.

K-Means minimises:

$$J = \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|_2^2 \tag{8}$$

Each fit used `n_init=20` and up to 500 iterations.

E. Classification: Match Outcome Prediction

1. Team-Level Aggregation

Individual composite scores were aggregated per (match, team): mean and minutes-weighted composite, positional sub-averages, top-3 individual composite mean, composite standard deviation, and five-match rolling team attack/defensive strength. Home/away vectors were pivoted to one row per match with prefixes `h_` and `a_`.

2. Differential Features

Fifteen differential features $\Delta f = f_{\text{home}} - f_{\text{away}}$ were computed, encoding the relative quality gap between sides.

3. Models and Evaluation

Logistic Regression optimises:

$$\mathscr{L}(\mathbf{W}) = -\sum_{i=1}^{N} \log p(y_i \mid \mathbf{x}_i; \mathbf{W}) + \frac{1}{C}\|\mathbf{W}\|_2^2 \tag{9}$$

with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected by `TimeSeriesSplit` grid search ($n = 3$). Class weights were balanced.

Random Forest Classification was tuned over $B \in \{100, 200\}$, max depth $\in \{5, 8, \text{None}\}$, min samples per leaf $\in \{5, 10, 20\}$, and max features $\in \{\sqrt{p}, 0.5p\}$ (full grids in Appendix VI.). Both models were evaluated on raw home/away features (29 features) and differential features (15 features). The primary metric is macro-averaged F_1.

IV. Results

A. Regression Results

1. Baseline and Ridge Regression

The position-mean baseline achieves $R^2 = 0.000$ on the test set. Ridge regression with enhanced features selected $\alpha = 200$ and achieved test $R^2 = 0.312$. Table 5 summarises all models.

Table 5: Regression Model Performance on Test Set (2024/25 Season)

Model	**Features**	**RMSE**	**MAE**	R^2
Baseline (pos. mean)	—	0.918	0.721	0.000
Ridge	Baseline	0.711	0.556	0.399
Ridge	Enhanced	0.696	0.544	0.421
Random Forest	Baseline	0.492	0.381	0.713
Random Forest	Enhanced	0.468	0.361	0.740
RF Grouped (mean)	Enhanced	0.328	0.253	0.947

2. Position-Grouped Random Forest and Feature Importance

The position-grouped RF with enhanced features was the best overall configuration (Figure 5). Test R^2: defender 0.966, winger 0.942, midfielder 0.941, forward 0.938 (mean 0.947).

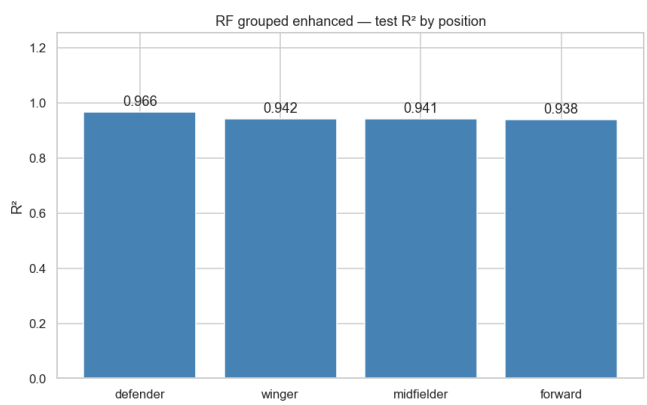

Figure 5: Test R^2 of position-grouped Random Forest (enhanced features). Defender achieves $R^2 = 0.966$; forward the lowest at 0.938.

Figure 6 shows the top 25 features by mean decrease in impurity. `roll5_defensive_actions` ranked first (importance ≈ 0.200), reflecting its value as a position-integrating proxy. `roll5_ShotsOnTarget` (≈ 0.120) and `roll5_touches` (≈ 0.107) ranked 2nd and 3rd. `roll5_assists` (0.094) and `roll5_goals` (0.074) ranked 4th and 5th, confirming the primacy of direct attacking output. The team-level feature `roll5_team_goals_scored` also appeared in the top 25, validating the enhanced feature set.

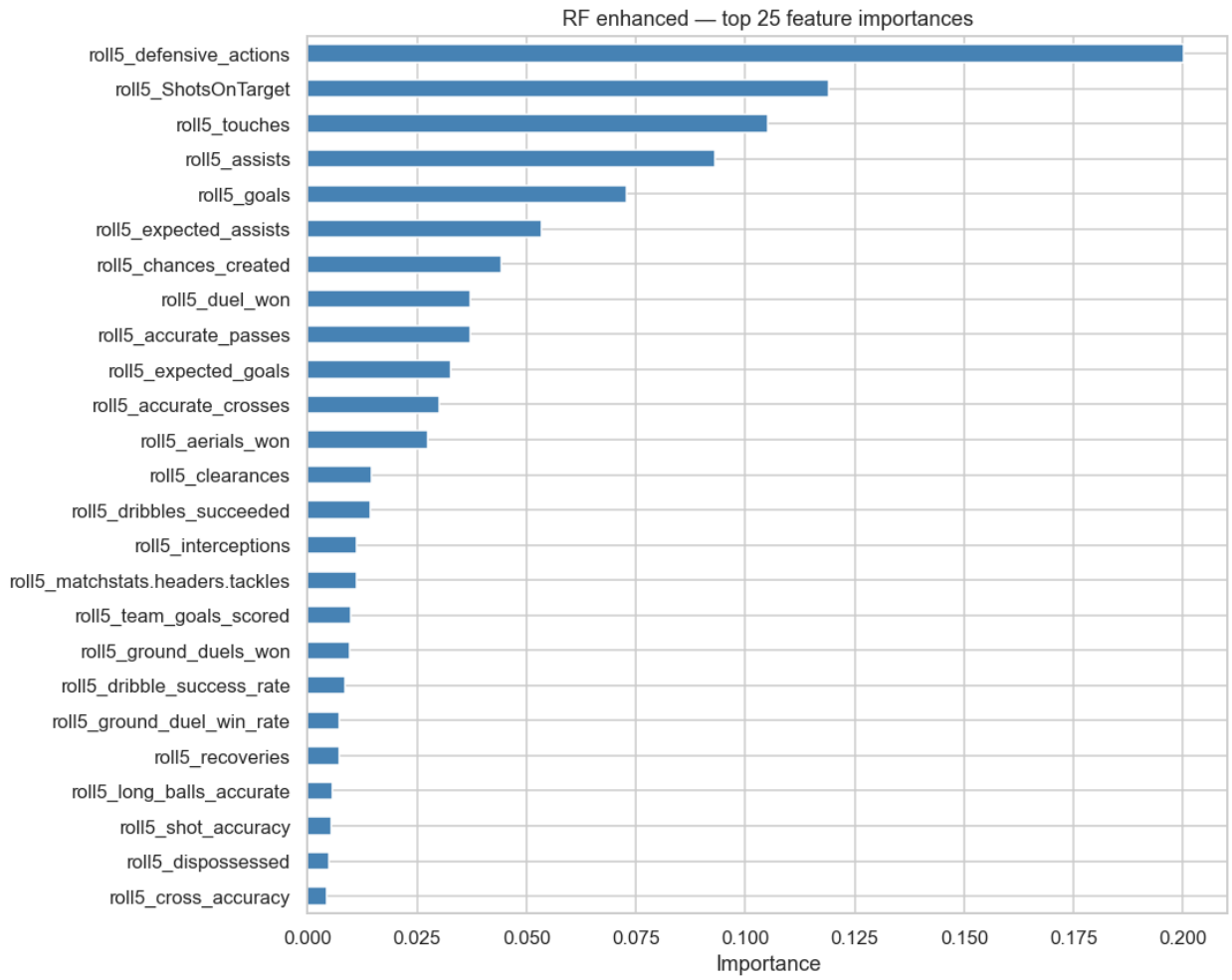

Figure 6: Top 25 features by mean decrease in impurity for the global Random Forest regression (enhanced features).

B. Clustering Results

1. PCA Variance and Model Selection

The cumulative explained variance curves per position group are provided in Appendix VI.. With 10 components, retained variance is 91–98% for outfield groups; 8 components retain 86% for goalkeepers.

Figure 7 shows the elbow and silhouette sweep. The silhouette peaks at $k = 2$ for all outfield groups (≈ 0.18–0.21); $k = 3$ is chosen for domain interpretability. The goalkeeper silhouette at $k = 2$ (≈ 0.63) far exceeds all outfield values, confirming a genuinely binary split.

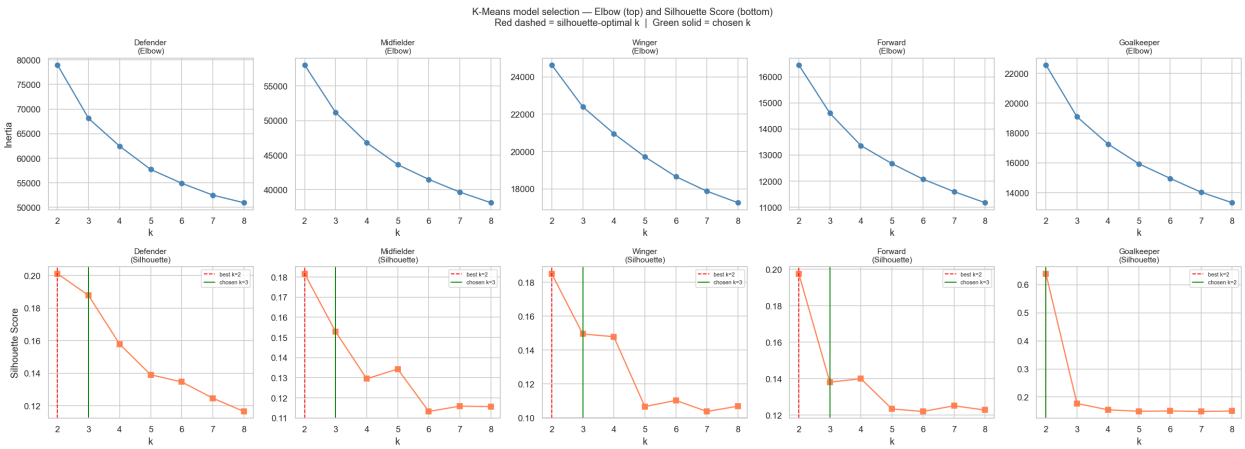

Figure 7: Inertia (top) and silhouette score (bottom) vs. k for all position groups. Red dashed: silhouette-optimal k; green solid: chosen k.

2. Archetype Profiles

Figure 8 shows the radar chart for defender archetypes. D0 (teal) dominates `aerials_won`, `headed_clearance`, and `shot_blocks`—a high-volume defensive CB. D1 (orange) leads on `interceptions` and `accurate_crosses`—a proactive attacking full-back. D2 (blue) is the compact defensive block profile. Radar charts for the remaining position groups (Midfielder, Winger, Forward, Goalkeeper) are provided in Appendix VI..

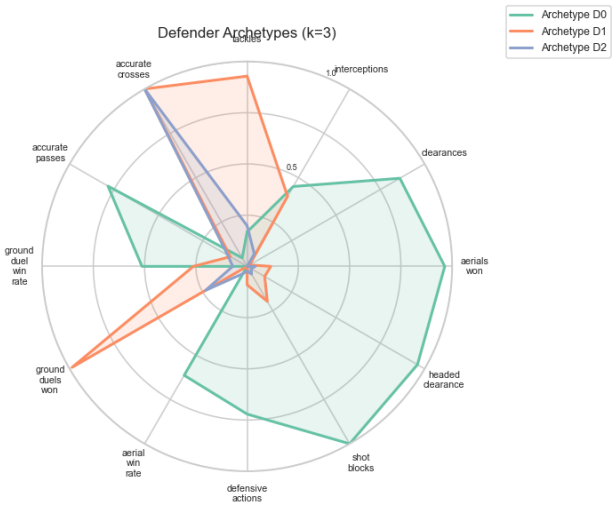

Figure 8: Defender archetypes ($k = 3$). D0 (teal) dominates `aerials_won`, `headed_clearance`, and `shot_blocks`—a high-volume defensive CB. D1 (orange) leads on `interceptions` and `accurate_crosses`—a proactive attacking full-back. D2 (blue) is the compact defensive block profile.

3. Archetype Heatmap

Figure 9 presents the cross-position heatmap of cluster means across ten universal features. MID–M0 dominates `goals` (1.81) and `assists` (1.29); DEF–D2 leads `expected_assists` (2.31) and `chances_created` (2.25), identifying creative ball-playing centre-backs; FOR–F2 ranks highest on `assists` (1.52) and `expected_assists` (1.81).

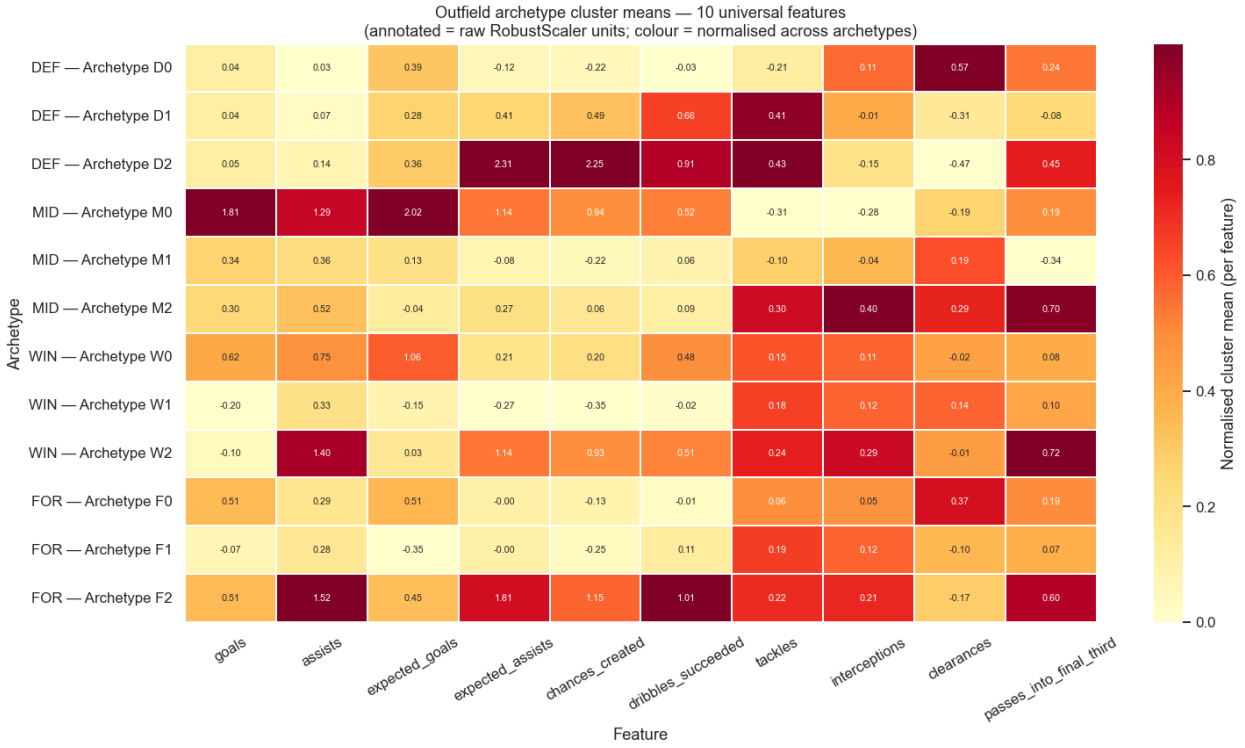

Figure 9: Outfield archetype cluster means for 10 universal features. Annotations are raw `RobustScaler` units; colour is normalised per feature column across archetypes.

Temporal stability and rating distributions per archetype are provided in Appendix VI..

4. PCA Scatter

The PCA 2D scatter coloured by archetype label for all five position groups is provided in Appendix VI.. The goalkeeper split is by far the most geometrically distinct; outfield archetypes overlap in PCA space, consistent with the moderate silhouette scores.

C. Classification Results

1. Cross-Validation and Test Performance

Tables 6 and 7 report cross-validation and test set performance respectively. Results are summarised in Table 6.

Table 6: Classification Cross-Validation (Training Seasons)

Model	**Features**	**CV Macro F_1**	**Best Params**
Logistic Regression	raw	0.381	$C = 0.1$, `lbfgs`
Logistic Regression	diff	0.368	$C = 0.01$, `lbfgs`
Random Forest	raw	0.542	depth=5, leaf=10
Random Forest	diff	0.517	depth=5, leaf=20

Table 7: Classification Test Set Performance (2024/25 Season)

Model	**Features**	**Accuracy**	**Macro F_1**
Baseline (majority class)	—	0.336	0.207
Logistic Regression	raw	0.456	0.371
Logistic Regression	diff	0.444	0.362
Random Forest	raw	**0.548**	**0.542**
Random Forest	diff	0.532	0.519

2. Confusion Matrix and Feature Importance

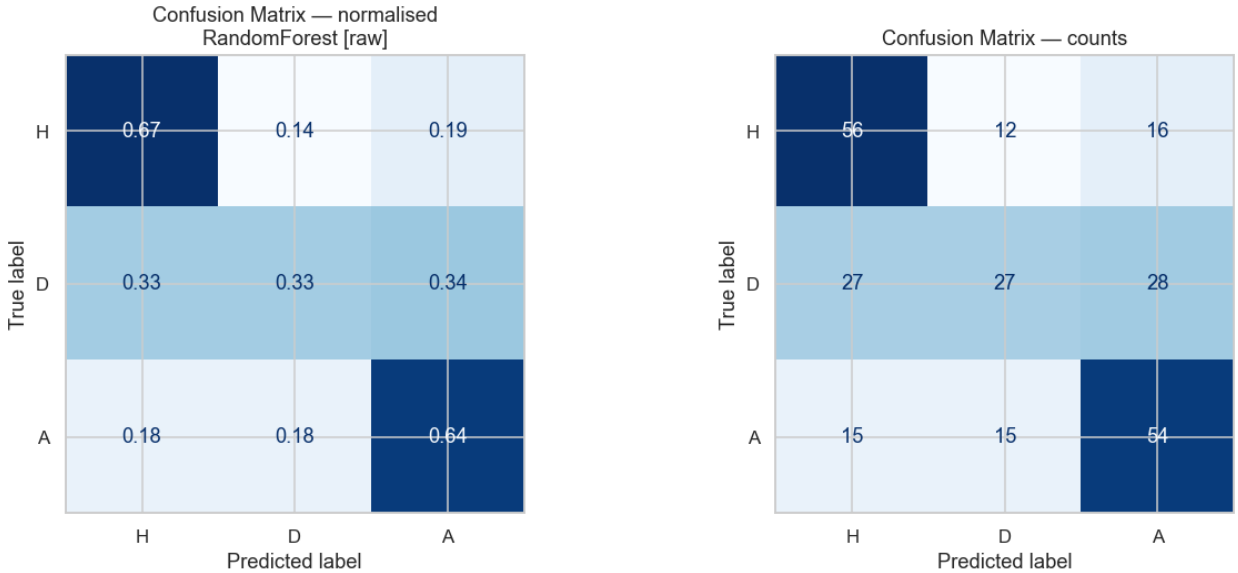

Figure 10: Confusion matrices for the best classifier on the 2024/25 test season. Left: normalised by true class; right: raw counts. H = home win, D = draw, A = away win.

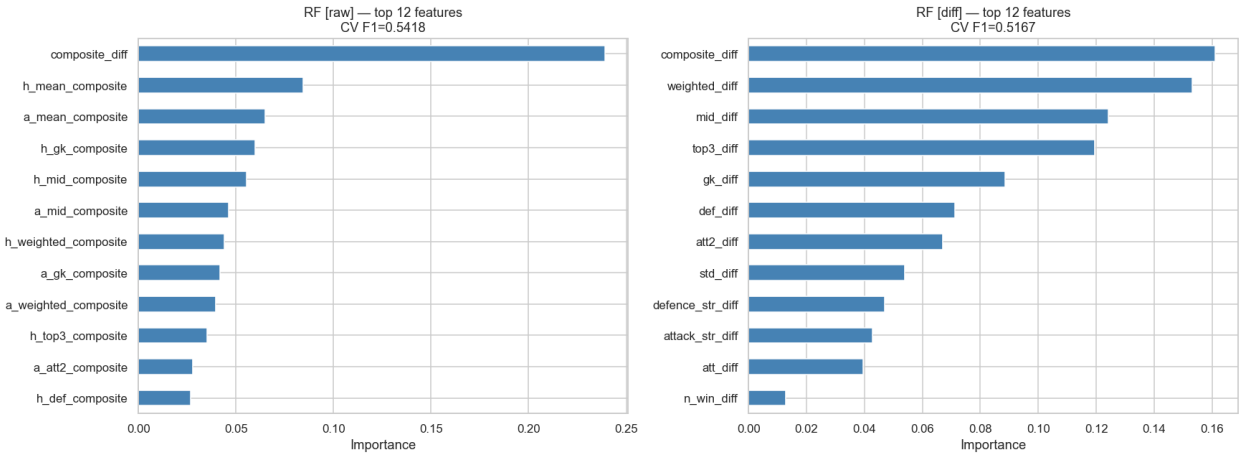

Figure 11: Feature importances (RF) or logistic regression coefficient heatmap for the best classification model.

V. Discussion

A. Interpretation of Results

1. Regression

The regression models demonstrated that five-match rolling statistics carry substantial predictive signal. Position-grouped Random Forest achieves test R^2 of 0.938–0.966, far above the baseline. The gap between Ridge and Random Forest confirms nonlinear interactions: a forward with high *xG and* high shot volume outperforms a linear combination.

The leading importance of `roll5_defensive_actions` (≈ 0.200) is initially counterintuitive—it reflects its utility as a position-integrating signal in a global model. However, the high R^2 values should be interpreted carefully: the composite score is itself a (near-)linear function of rolling features, so a flexible nonlinear model can approximate it closely. The more meaningful evaluation is the downstream classification performance.

2. Clustering

K-Means recovered tactically coherent groupings despite moderate silhouette scores (0.13–0.21 outfield). The goalkeeper split was the cleanest (≈ 0.63). The heatmap in Figure 9 confirms expected structure: MID–M0 is the highest-producing midfielder; DEF–D2 paradoxically leads chance creation (expected assists 2.31, chances created 2.25), identifying ball-playing centre-backs whose progressive contributions dominate the universal feature set. Temporal stability (Appendix VI.) confirms archetypes are not season-specific artefacts.

3. Classification

The Random Forest classifier with raw home/away features achieved the best test performance: accuracy 0.548 and macro $F_1 = 0.542$, compared to a majority-class baseline of $F_1 = 0.207$—a gain of +0.335. The raw feature set (29 features, home and away separately) outperformed the differential-only set ($F_1 = 0.519$), suggesting the classifier extracts additional signal from absolute team quality levels rather than purely the relative gap. The draw class (D) showed recall of only 0.33 versus 0.67 for home wins and 0.64 for away wins (Figure 10), a well-known

challenge in football outcome modelling [7]. The dominant feature across both RF variants was `composite_diff` (≈ 0.24 importance, raw; ≈ 0.16, diff), confirming that the overall composite gap between home and away squads is the strongest single predictor available from the pipeline.

B. Why Certain Methods Performed Better

Random Forest outperformed Ridge regression at the regression stage because the composite score is not globally linear in its rolling inputs. At the classification stage, the raw feature set (CV $F_1 = 0.542$) outperformed the differential-only set ($F_1 = 0.517$), which is counterintuitive given that differentials encode the most direct signal of the quality gap. Feature importance analysis (Figure 11) reveals why: in the raw model, `composite_diff` (which already appears as a feature) still dominates at 0.24, but the home-team composite values contribute independent signal—the absolute strength of the home side matters beyond the relative gap, likely because home advantage interacts with team quality in a way that differences alone cannot capture. Position-grouped regression models outperformed the global model for most positions, confirming that feature-importance structures differ sufficiently across roles to justify dedicated fits.

C. Lessons Learned

The temporal split was genuinely harder than a size-equivalent random split: the 2024/25 test season includes player movements, tactical innovations, and club changes absent from 2021/22. The near-zero GK composite correlation with `rating_title` ($r = -0.062$) is a useful reminder that goalkeeper evaluation from volume statistics alone is unreliable. The choice of $k = 3$ for outfield clustering over the silhouette-optimal $k = 2$ is an acknowledged trade-off between statistical optimality and domain interpretability.

D. Future Improvements

Natural extensions include: (i) incorporating opponent xG as a fixture difficulty prior; (ii) extending clustering to team-level tactical archetypes; (iii) adding Elo-style ratings as classification priors; and (iv) calibrating classification probabilities for live match-preview use.

VI. Conclusions

This paper presented GOALS, a machine learning pipeline for Premier League match outcome prediction built on four complete seasons of player-level data. Three modelling tasks were integrated into a coherent end-to-end system.

The regression stage achieved test R^2 of 0.938–0.966 across position-grouped Random Forest models (mean 0.947), with `roll5_defensive_actions` emerging as the dominant feature. The clustering stage discovered interpretable, temporally stable archetypes; goalkeeper clustering produced the cleanest binary separation (silhouette ≈ 0.63) while outfield groups showed expected moderate scores (0.13–0.21). The classification stage achieved test accuracy of 0.548 and macro $F_1 = 0.542$ (Random Forest, raw features), a gain of +0.335 over the majority-class baseline; draws remained the hardest class to predict (recall 0.33 vs. 0.67 for home wins).

Key insights include the necessity of temporal data splitting, the value of position-specific modelling, and the trade-off between statistical optimality and domain interpretability in cluster number selection. The pipeline is fully reproducible and designed for incremental extension as new season data arrives.

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VITA

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Appendix A: Supplementary EDA Figures

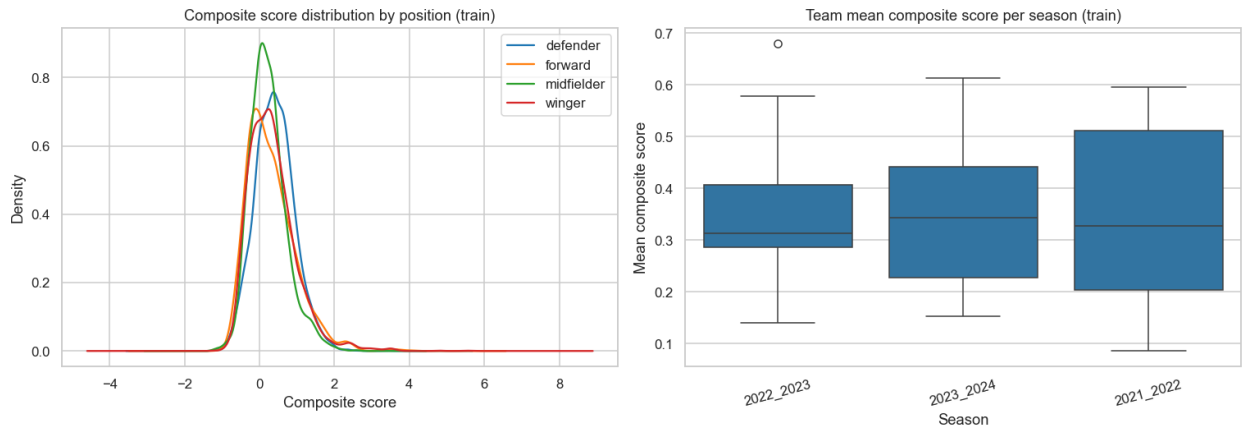

Figure 12: Left: composite score KDE by position group. Right: team mean composite per season, confirming consistency across training seasons.

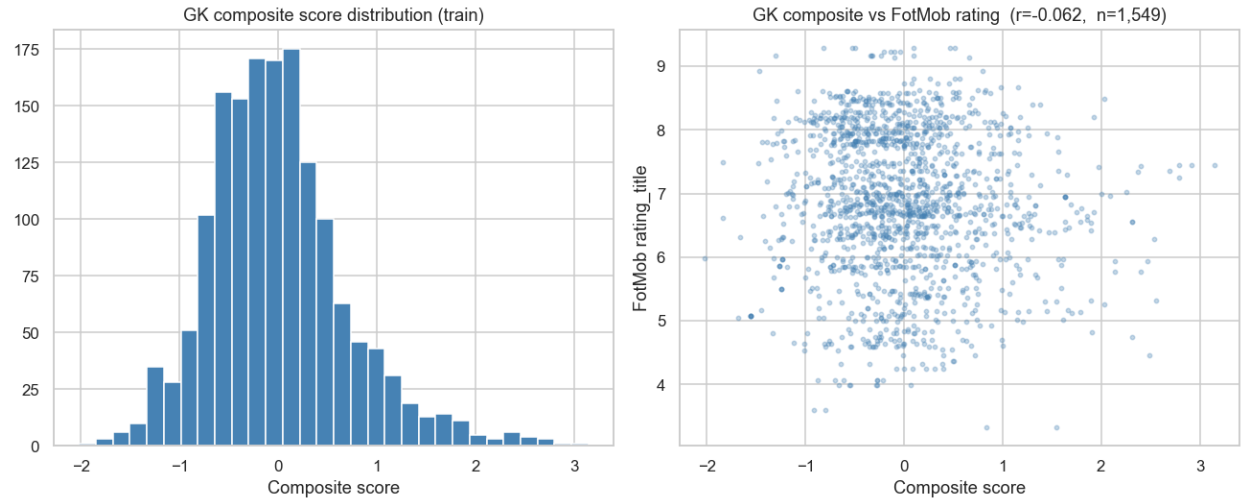

Figure 13: GK composite score distribution (left) and scatter vs. FotMob rating (right, $r = -0.062$, $n = 1{,}549$). The near-zero correlation is expected for goalkeepers (see text).

Appendix B: Clustering Supplementary Figures

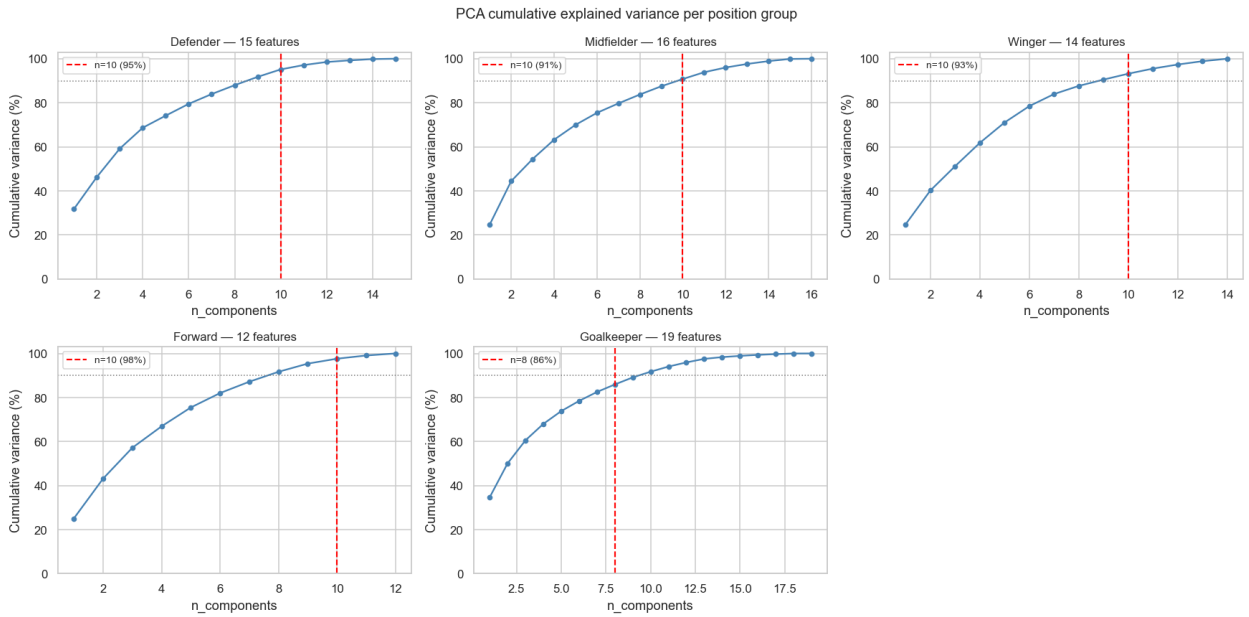

Figure 14: Cumulative explained variance vs. n PCA components per position group. Red dashed: chosen n; grey dotted: 90% threshold.

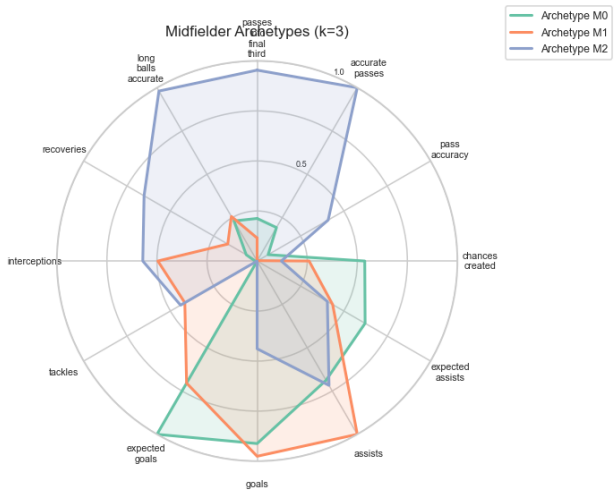

Figure 15: Midfielder archetypes ($k = 3$). M2 (blue) dominates `accurate_passes` and `pass_accuracy`—a deep-lying distributor. M0 (teal) and M1 (orange) lead on goal and assist metrics—box-to-box and creative profiles.

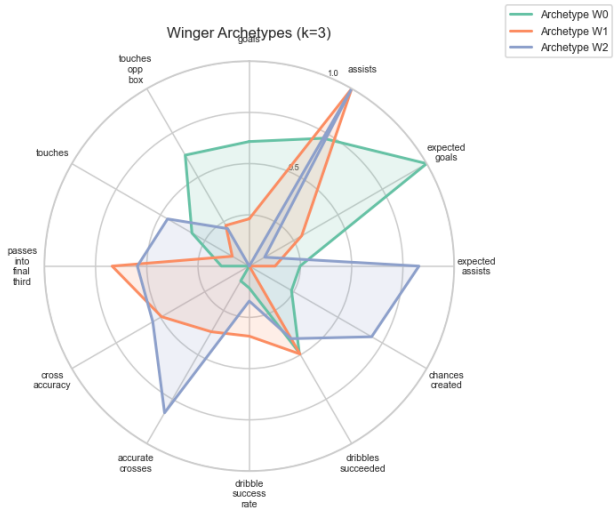

Figure 16: Winger archetypes ($k = 3$). W0 (teal) leads on `assists` and `expected_goals`—an inverted threat carrier. W2 (blue) dominates `expected_assists` and `chances_created`—a wide playmaker. W1 is the supporting/low-event profile.

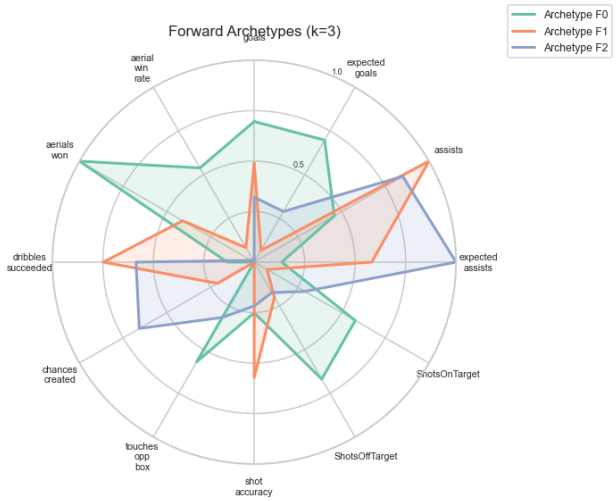

Figure 17: Forward archetypes ($k = 3$). F0 (teal) leads on `expected_goals` and `ShotsOnTarget`—a clinical striker. F2 (blue) dominates `expected_assists` and `assists`—a link-up forward. F1 (orange) shows higher `aerials_won`—a target-man profile.

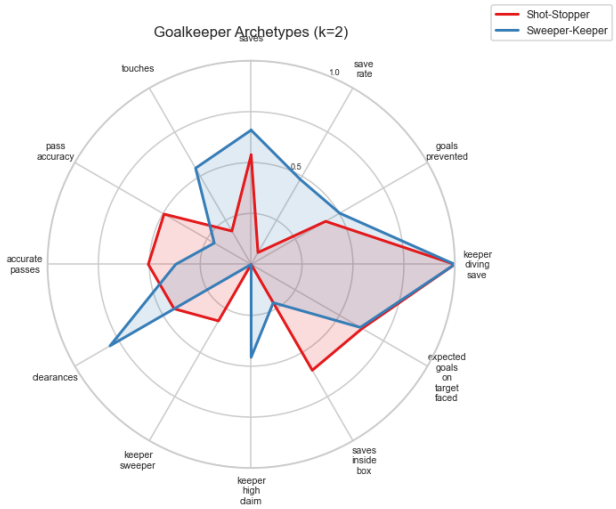

Figure 18: Goalkeeper archetypes ($k = 2$). The Shot-Stopper (red) leads on `saves`, `keeper_diving_save`, and `saves_inside_box`. The Sweeper-Keeper (blue) leads on `keeper_sweeper`, `pass_accuracy`, and `touches`.

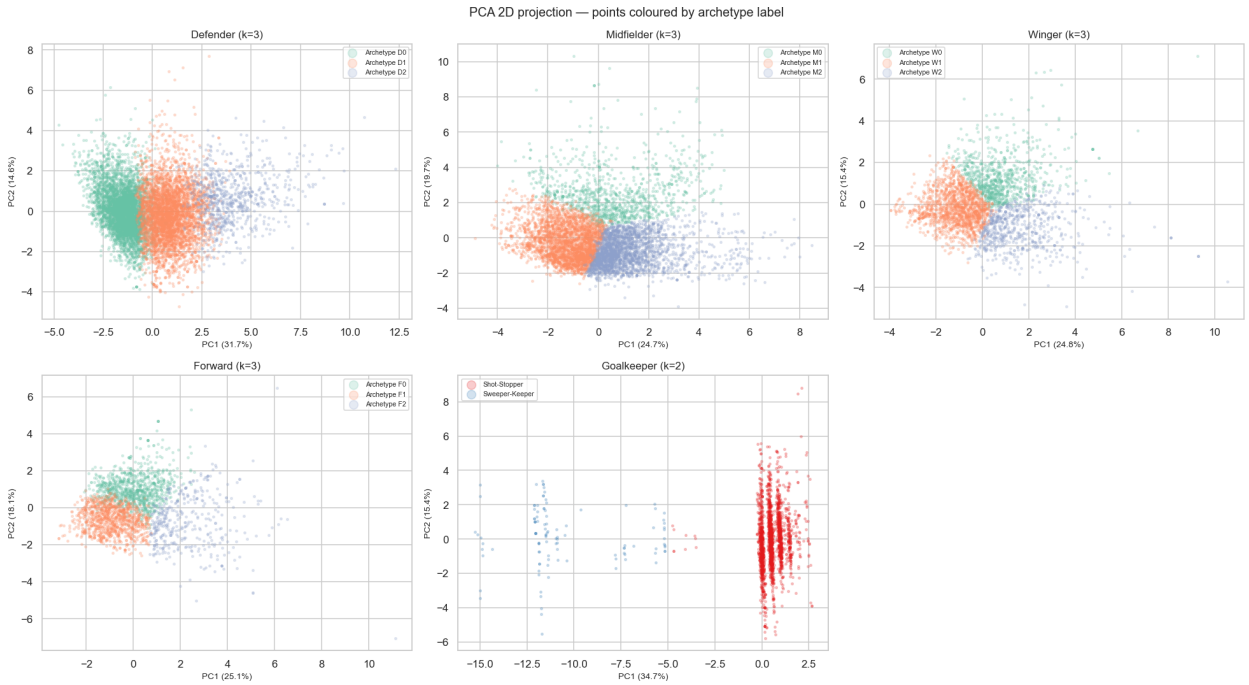

Figure 19: PCA 2D projection coloured by archetype label for all five position groups. The goalkeeper split (bottom-right) is the most distinct.

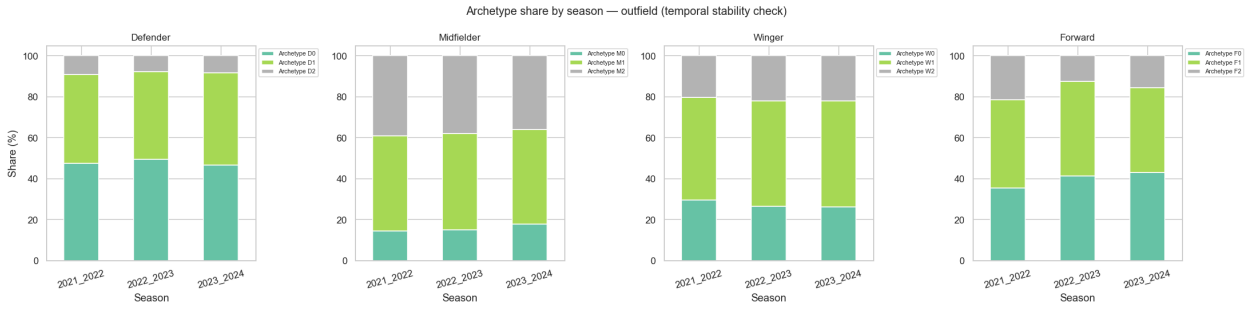

Figure 20: Archetype proportion by season for each outfield position group, confirming temporal stability across 2021/22–2023/24.

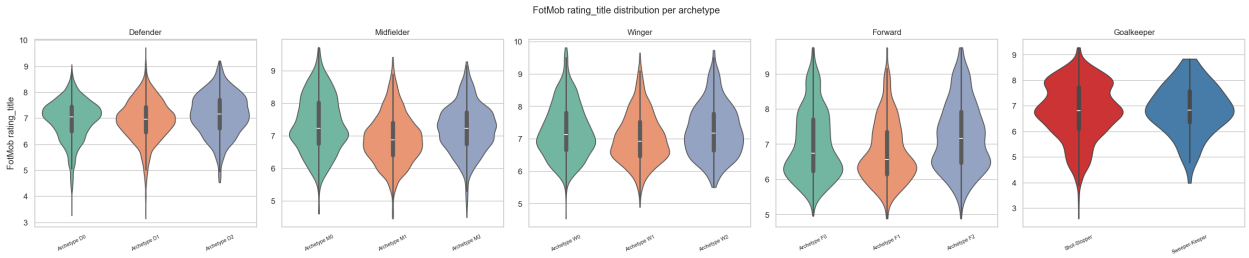

Figure 21: FotMob `rating_title` distribution per archetype and position group. Inner box marks IQR and median.

Appendix C: Hyperparameter Search Grids

Table 8: Regression Hyperparameter Grids

Model	**Parameter**	**Values**
Ridge	α	$\{0.01, 0.1, 1, 10, 100, 500\}$
	CV	neg. RMSE, `TimeSeriesSplit`$(n = 5)$
Random Forest	B	$\{100, 200, 300\}$
	max depth	$\{4, 6, 8, 10, \text{None}\}$
	min samples leaf	`randint`$(5, 30)$
	max features	$\{\sqrt{p}, 0.5p\}$

Table 9: Classification Hyperparameter Grids

Model	**Parameter**	**Values**
Logistic Reg.	C	$\{0.01, 0.1, 1, 10, 100\}$
	solver	{`lbfgs`, `saga`}
Random Forest	B	$\{100, 200\}$
	max depth	$\{5, 8, \text{None}\}$
	min samples leaf	$\{5, 10, 20\}$
	max features	$\{\sqrt{p}, 0.5p\}$

Appendix D: Key Code Listings

```
def compute_composite(df, weights_by_pos, z_params=None):
    """Position-aware composite: z-score normalised weighted sum."""
    df = df.copy()
    scores  = pd.Series(np.nan, index=df.index)
    all_cols = {c for cols in weights_by_pos.values() for c in cols}
    if z_params is None:
        z_params = {col: (df[col].mean(), df[col].std(ddof=0))
                    for col in all_cols if col in df.columns}
    for pos, weights in weights_by_pos.items():
        mask  = df['position_group'] == pos
        score = pd.Series(0.0, index=df[mask].index)
        for col, w in weights.items():
            if col not in df.columns:
                continue
            mu, sigma = z_params.get(col, (0.0, 1.0))
            z = (df.loc[mask, col] - mu) / (sigma or 1.0)
            score += w * z
        scores[mask] = score
    df['composite_score'] = scores
```

```
    return df, z_params
```

Listing 1: Composite score computation (Notebook 03).

```
def add_rolling_features(df, roll_cols, window=5):
    """shift(1).rolling(window).mean() per player, by match_date."""
    df = df.sort_values(['player_id', 'match_date']).copy()
    present = [c for c in roll_cols if c in df.columns]
    rolled = (
        df.groupby('player_id')[present]
          .apply(lambda x: x.shift(1).rolling(window,
                                              min_periods=1).mean())
          .reset_index(level=0, drop=True)
    )
    rolled.columns = [f'roll5_{c}' for c in present]
    df = pd.concat([df, rolled], axis=1)
    return df.dropna(subset=rolled.columns.tolist(),
                     how='all').reset_index(drop=True)
```

Listing 2: Rolling feature engineering (Notebook 03).